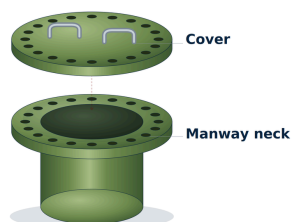


FRP Manways & Covers

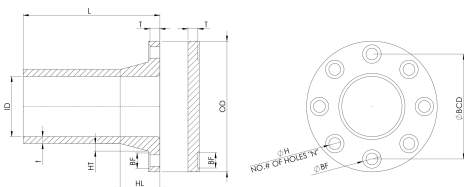


20 – 36 in. SIZE RANGE

0.5 • 15 • 25 • 75 PSI CLASSES

Set or single NECK AND COVER

Manway necks and matching covers for FRP tanks and vessels. Cover thickness equals flange thickness (T) for the corresponding pressure rating. Necks and covers are supplied as a matched set or as individual items.



ID nominal inside diameter
T flange & cover thickness
t stub wall thickness
HL hub length
BCD bolt circle diameter
BF back-face diameter

OD flange outside diameter
L overall stub length
HT hub thickness at root
N number of bolt holes
H bolt hole diameter
P pressure rating

ID nominal dia.	P pressure	OD flange dia.	T flange & cover thk.	L stub length	t wall thk.	HT hub thk.	HL hub length	N holes	BCD bolt circle	H hole dia.	BF back-face
in	psi	in	in	in	in	in	in	#	in	in	in
20	0.5	27-5/8	3/8	12	1/4	3/16	1-1/2	20	25	1/2	15/16
20	15	27-5/8	1	12	3/8	1/2	4	20	25	3/4	1-1/2
20	75	27-5/8	1-5/8	12	1/2	13/16	6-1/2	20	25	1-1/4	2-1/2
24	0.5	32-1/8	3/8	12	1/4	3/16	1-1/2	20	29-1/2	1/2	15/16
24	15	32-1/8	1-1/4	12	3/8	5/8	5	20	29-1/2	3/4	1-1/2
24	75	32-1/8	1-7/8	12	5/8	15/16	7-1/2	20	29-1/2	1-3/8	2-3/4
30	25	38-7/8	1-3/8	12	5/16	11/16	5-1/2	28	36	1-3/8	2-3/4
36	25	46-1/8	1-3/4	12	3/8	7/8	7	32	42-3/4	1-5/8	3-1/4

NOTES

- Special drilling compatible with ASME RTP-1 on 15 psi manways and covers.
- Class 150 drilling (ASME B16.5) on 75 psi manways and covers.
- ASME B16.1 Class 125 drilling on 30 in. and 36 in. 25 psi manways and covers.
- The 0.5 psi range uses the RTP-1 bolt circle with reduced 1/2 in. bolt holes.

TOLERANCES

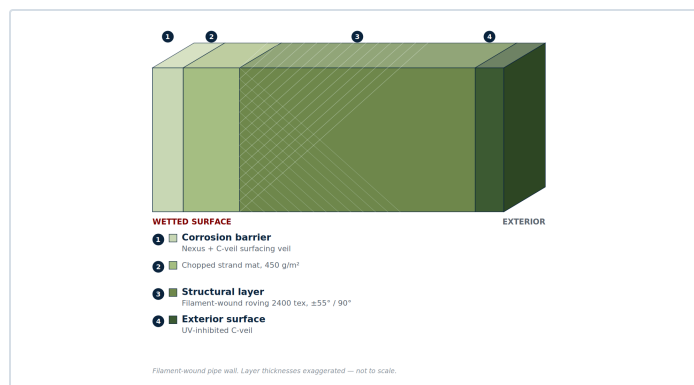
- BCD: ± 0.06 in. • Hole spacing: ± 0.03 in. • Eccentricity: ± 0.06 in. (1.5 mm).
- Flange and cover face flatness: $\pm 1/32$ in.
- Back-face (BF) dimensions suit ASME B18.22.1 Type A Narrow washers.
- All thicknesses include the corrosion barrier.



Resins, Laminate & Standards

Every component is built as a layered laminate. The layer names below are those used by ASTM C582 and ASME RTP-1 and are the common vocabulary of the industry. What differs between products is not the structure but the ply build-up and the way the structural layer is laid down. Wall thicknesses quoted throughout this catalogue are total thicknesses and include the corrosion barrier.

LAMINATE CONSTRUCTION



Filament-wound pipe wall. Layer thicknesses not to scale.

Inner surface

A resin-rich layer reinforced with a surfacing veil — glass, synthetic or carbon. Nominally 0.010–0.020 in. This is the surface the process fluid touches.

Interior layer

Chopped strand mat, two plies minimum. With the inner surface it forms the **corrosion-resistant barrier** — the part of the wall that does the chemical work, and the part excluded from the strength calculation.

Structural layer

Carries the pressure and the mechanical load. Filament wound on pipe; contact moulded — hand lay-up — on every fitting. The first ply against the barrier is always mat.

Outer surface

A resin coat, paraffinated for a full surface cure. UV-inhibited exterior standard on pipe, available on fittings.

Ply-by-ply build-up differs by product and method. The filament-wound pipe schedule is tabulated on page 07; fitting schedules are held by the manufacturer — see the scope note below.

STANDARDS OF MANUFACTURE AND REFERENCE

CGSB 41-GP-22	Process equipment — reinforced polyester, glass fibre. Governing Canadian fabrication standard; covers tanks, pipe and ductwork, and establishes the 10:1 safety factor.
ASTM D5421	Contact-Molded "Fiberglass" Flanges. Governs the stub flanges and manway flanges.
ASME RTP-1	Referenced for bolt-hole drilling compatibility with RTP-1 vessel nozzle and manway patterns only — see the scope note below.
ASME B16.1 Cl. 125	Bolt-hole drilling, NPS 30 and 36.
ASME B18.22.1	Type A Narrow plain washers — governs back-face (BF) diameters.

RESIN SYSTEMS

Derakane 411-350

Bisphenol-A epoxy vinyl ester

HDT	105 °C / 220 °F	TENSILE	86 MPa / 12,000 psi
MODULUS	3,200 MPa	ELONGATION	5–6 %

The standard offering. Broad resistance to acids, alkalis, bleaches and solvents.

Derakane 470-300

Epoxy novolac vinyl ester

HDT	150 °C / 300 °F	TENSILE	85 MPa / 12,500 psi
MODULUS	3,600 MPa	ELONGATION	3–4 %

Specified where elevated temperature, solvent or oxidiser service (chlorine, flue gas) demands higher thermal and chemical resistance.

HDT is a material property, not a service rating. Maximum service temperature is governed by the chemical environment — confirm against the resin manufacturer's chemical resistance guide before specifying.

DESIGN BASIS

- Minimum **10:1 safety factor** on the structural laminate, per CGSB 41-GP-22.
- The corrosion-resistant barrier is excluded from the strength calculation — the safety factor applies to the structural wall only.
- Filament-wound structural layers are designed on a **2:1 hoop-to-axial** stress ratio.

GASKETS & BOLTING

- 1/8 in. full-face elastomeric gasket, Shore A60 ± 5, minimum design seating stress (y) 100 psi, gasket factor (m) 1.
- Flat-face mating flanges only — never bolt FRP to a raised-face metallic flange without a full-face adaptor.
- Tighten in a star pattern in equal stages. FRP flanges are damaged by over-torquing — ask us for torque values.

STANDARDS SCOPE

- ASME RTP-1 is a vessel standard** rated to 15 psig. It does not cover piping or pipe fittings. Where we cite it, we mean bolt-hole drilling compatibility with RTP-1 vessel nozzle and manway patterns — nothing here is offered as an RTP-1 certified vessel component.
- We are a stockist. Item-specific laminate schedules and conformance documentation come from the manufacturer against a specific order — ask at enquiry stage.



FRP DEPOTS

CANADA'S DIRECT SOURCE FOR FRP COMPONENTS

How to order

Send your enquiry to sales@frpdepots.com or call **613-499-4950**. Include the following and we will quote from stock where we can.

- | | |
|---|---|
| 01 Product type and nominal diameter | 02 Pressure rating required |
| 03 Resin system — D411 or D470 | 04 Quantity and length (pipe, couplings) |
| 05 Any non-standard drilling, stub length or joint preparation | 06 Drawings or job specification, if you have them |

Full dimensional data, tolerances and design basis are published in **FRPD-SPEC-001 Revision R3**. Request the current revision with your enquiry.

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